analysis

Import data

```
dataset_restaurants <- read_csv(here("data", "restaurants_attributes_extracted.csv"))
```

Exploring star rating

```
#Summary statistics
summary(dataset_restaurants$stars)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   1.000   3.000   3.500   3.465   4.000   5.000
```

```
#Mean and SD
m_stars <- mean(dataset_restaurants$stars, na.rm = TRUE)
sd_stars <- sd(dataset_restaurants$stars, na.rm=TRUE)

m_stars
```

```
## [1] 3.464931
```

```
sd_stars
```

```
## [1] 0.8088196
```

```
#Histogram
plot_stars <- ggplot(dataset_restaurants, aes(x = stars)) +
  geom_histogram(binwidth = 0.5, fill = "#A6CEE3", color = "white") +
  labs(
    title = "Distribution of Restaurant Star Ratings",
    x = "Star Rating",
    y = "Number of Restaurants"
  ) +
  theme_minimal()

plot_stars
```

```
ggsave(
  filename = here("gen", "output", "plot_stars.png"),
  plot = plot_stars,
  width = 7,
  height = 5
)
```

The figure shows the distribution of restaurant star ratings. Ratings range from 1 to 5 stars, with most restaurants clustered around ratings between approximately 3 and 4.5 stars. The distribution is slightly left-skewed, indicating that relatively few restaurants receive very low ratings.

The average rating in the data set is 3.46 (SD = 0.81), suggesting that overall customer evaluations are generally positive. Since star rating is the dependent variable in our analysis, understanding its distribution provides important context for interpreting the effect of delivery options, dine-experience and price range in later sections.

Exploring price range

```
#Summary statistics

summary(dataset_restaurants$price_range)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   1.000   1.000   2.000   1.608   2.000   4.000
```

```r
#Histogram

plot_price <- ggplot(dataset_restaurants, aes(x = factor(price_range))) +
  geom_bar(fill = "#A6CEE3") +
  labs(
    title = "Distribution of Restaurant Price Range",
    x = "Price Range",
    y = "Number of Restaurants"
  ) +
  theme_minimal()

plot_price
```

```r
ggsave(
  filename = here("gen", "output", "plot_price_range.png"),
  plot = plot_price,
  width = 7,
  height = 5
)
```

The figure shows the distribution of restaurants across the four price categories. The majority of restaurants fall within the lower price segments (price levels 1 and 2), with price level 2 being the most common category. In contrast, relatively fewer restaurants are classified as higher-priced establishments (levels 3 and 4), with 4 only representing a very small proportion of the sample.

This indicates that the data set is skewed toward lower priced restaurants. Since price range serves as a moderating variable in the analysis, the distribution should be considered when interpreting interaction effects in later sections

Cluster score

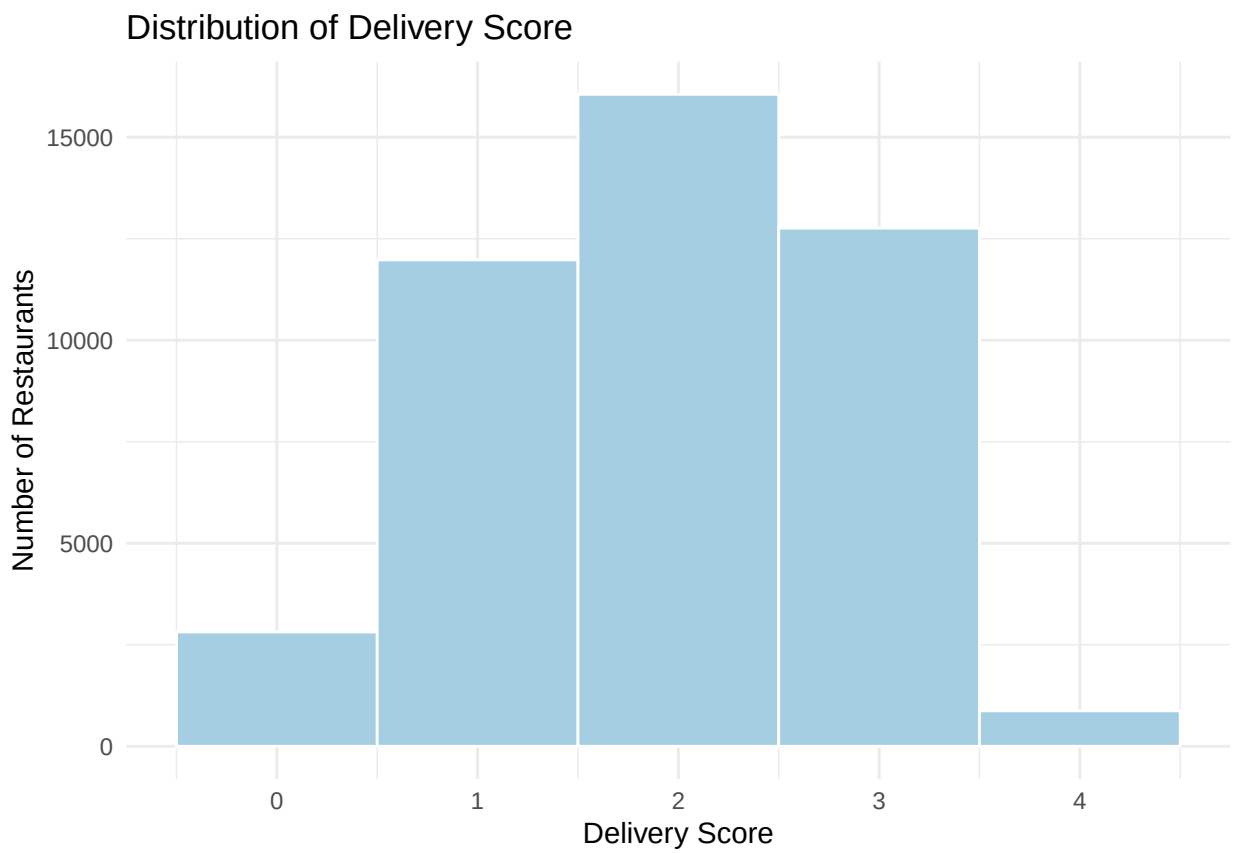
```r
#Summary Statistics
summary(dataset_restaurants$delivery_score)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    0.00    1.00    2.00    1.93    3.00    4.00
```

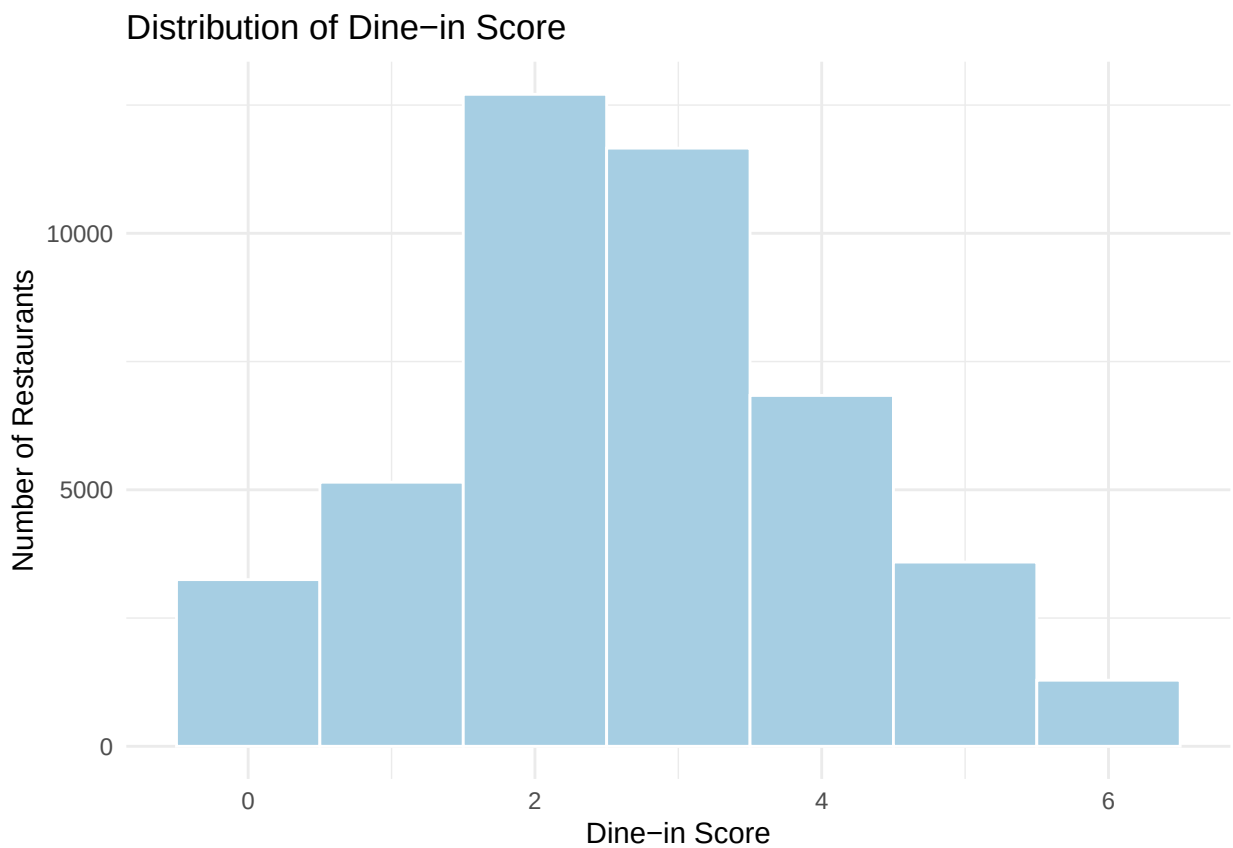
```r
summary(dataset_restaurants$dinein_score)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   0.000   2.000   3.000   2.666   4.000   6.000
```

```r
#Distribution of the cluster score
plot_delivery_hist <- ggplot(dataset_restaurants, aes(x = delivery_score)) +
  geom_histogram(binwidth = 1, fill = "#A6CEE3", color = "white") +
  labs(
    title = "Distribution of Delivery Score",
    x = "Delivery Score",
    y = "Number of Restaurants"
  ) +
  theme_minimal()

plot_delivery_hist
```




```
ggsave(
  filename = here("gen", "output", "plot_delivery_hist.png"),
  plot = plot_delivery_hist,
  width = 7,
  height = 5
)

plot_dinein_hist <- ggplot(dataset_restaurants, aes(x = dinein_score)) +
  geom_histogram(binwidth = 1, fill = "#A6CEE3", color = "white") +
  labs(
    title = "Distribution of Dine-in Score",
    x = "Dine-in Score",
    y = "Number of Restaurants"
  ) +
  theme_minimal()

plot_dinein_hist
```




```r
ggsave(
  filename = here("gen", "output", "plot_dinein_hist.png"),
  plot = plot_dinein_hist,
  width = 7,
  height = 5
)
#Scatterplot for Delivery_score
plot_delivery_stars <- ggplot(dataset_restaurants, aes(x = delivery_score, y = stars)) +
  geom_point(alpha = 0.2) +
  geom_smooth(method = "lm", color = "black") +
  labs(
    title = "Delivery Score and Star Ratings",
    x = "Delivery Score",
    y = "Star Rating"
  ) +
  scale_y_continuous(limits = c(1, 5)) +
  theme_minimal()

plot_delivery_stars
```

```r
ggsave(
  filename = here("gen", "output", "plot_delivery_stars.png"),
  plot = plot_delivery_stars,
  width = 7,
  height = 5
)
```

```r
#Delivery score
  model_delivery <- lm(stars ~ delivery_score, data = dataset_restaurants)
#Summary of the delivery score
summary(model_delivery)
```

```
##
## Call:
## lm(formula = stars ~ delivery_score, data = dataset_restaurants)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.58689 -0.46053  0.03947  0.53947  1.66584
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)     3.586892   0.008738  410.50   <2e-16 ***
## delivery_score -0.063183   0.004070  -15.52   <2e-16 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8066 on 44482 degrees of freedom
## Multiple R-squared:  0.005388,   Adjusted R-squared:  0.005366
## F-statistic:   241 on 1 and 44482 DF,  p-value: < 2.2e-16
```

The delivery and dine-in scores take discrete values, as they are constructed from multiple binary service indicators. The majority of restaurants fall within the lower-to-middle range of both scores, indicating that only a subset of establishments offers many delivery or dine-in related services.

A simple linear regression analysis shows a statistically significant negative relationship between delivery score and star rating. Restaurants offering more delivery-related services tend to receive slightly lower ratings.

However, the effect size is small. A one-point increase in delivery score is associated with a decrease of only 0.063 stars. Even across the full delivery scale, the maximum predicted difference in rating is approximately 0.25 stars.

Furthermore, delivery score explains less than 1% of the variance in star ratings, indicating that its pratical importance is limited.

Testing Moderation effect with delivery_score

```r
#Moderation effect
model_moderation <- lm(stars ~ delivery_score * price_range,
                       data = dataset_restaurants)
#Summary of moderation
summary(model_moderation)
```

```
##
## Call:
## lm(formula = stars ~ delivery_score * price_range, data = dataset_restaurants)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.76534 -0.56163  0.04022  0.52573  1.83487
##
## Coefficients:
##                            Estimate Std. Error t value Pr(>|t|)
## (Intercept)                3.673507   0.023721 154.866  < 2e-16 ***
## delivery_score            -0.315587   0.011135 -28.341  < 2e-16 ***
## price_range               -0.066411   0.012885  -5.154 2.56e-07 ***
## delivery_score:price_range  0.168266   0.006347  26.509  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7897 on 44480 degrees of freedom
## Multiple R-squared:  0.0468, Adjusted R-squared:  0.04673
## F-statistic: 727.9 on 3 and 44480 DF,  p-value: < 2.2e-16
```

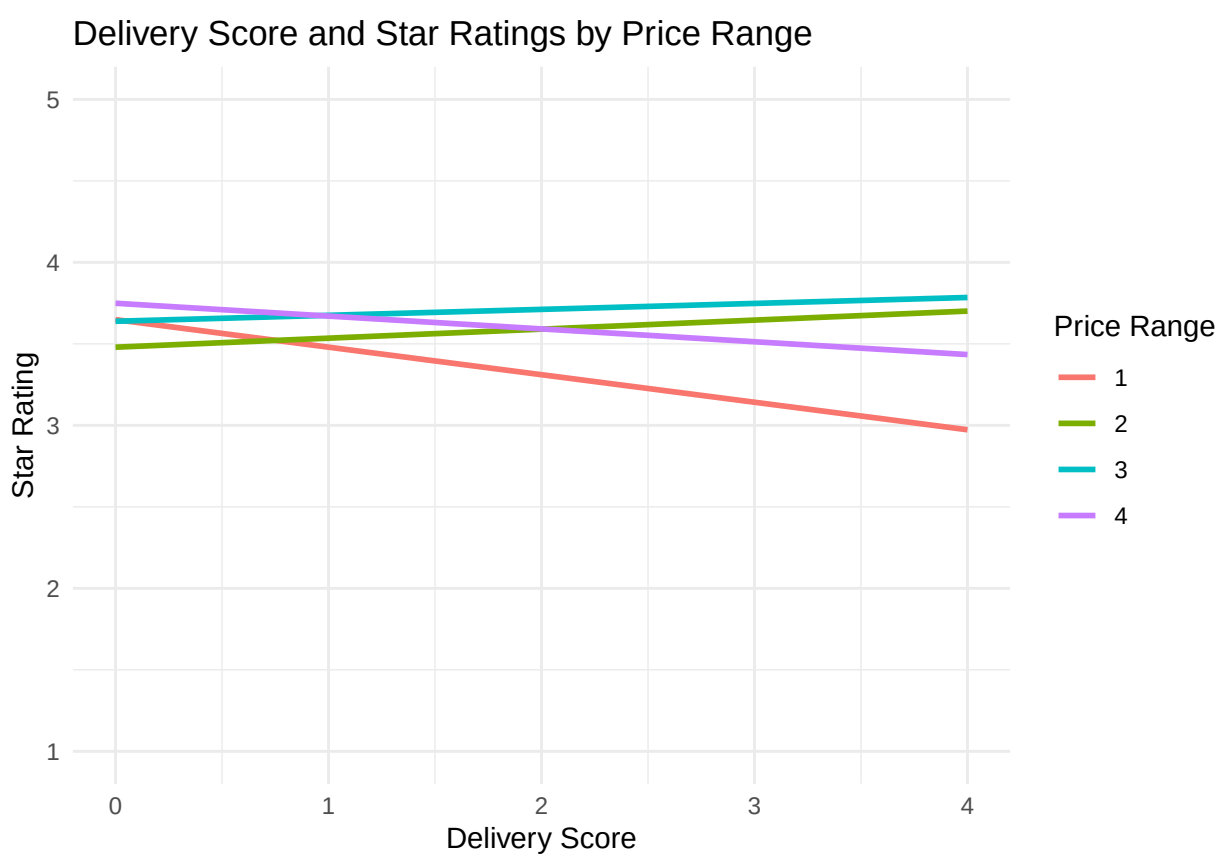
A regression model including an interaction between delivery score and price range reveals a positive and statistically significant interaction effect (B = 0,168, p < 001). This indicates that price range moderates the relationship between delivery services and star ratings.

Specifically, the negative association between delivery score and ratings becomes weaker as price range increases and eventually returns positive for higher-priced restaurants. For lower-priced establishments,

offering more delivery-related services is associated with slightly lower ratings. In contrast, for higher-priced restaurants, delivery services are associated with higher ratings.

The inclusion of the interaction improves model fit (R^2 = 0.047), compared to the model without moderation (R^2 = 0.005). This suggests that price range plays a meaningful moderating role in explaining variation in restaurant ratings

```
#Data plot for delivery_score
plot_delivery_mod <- ggplot(
  dataset_restaurants,
  aes(x = delivery_score, y = stars, color = factor(price_range))
) +
  geom_smooth(method = "lm", se = FALSE) +
  scale_y_continuous(limits = c(1, 5)) +
  labs(
    title = "Delivery Score and Star Ratings by Price Range",
    x = "Delivery Score",
    y = "Star Rating",
    color = "Price Range"
  ) +
  theme_minimal()

plot_delivery_mod
```




```
ggsave(
  filename = here("gen", "output", "plot_delivery_mod.png"),
```

```
  plot = plot_delivery_mod,
  width = 7,
  height = 5
)
```

This figure displays the moderating role of price range on the relationship between delivery score and star ratings. For lower-priced restaurants, an increase in delivery-related services is associated with lower customer ratings, as indicated by the negative slope. As price range increases, this relationship weakens and gradually becomes positive.

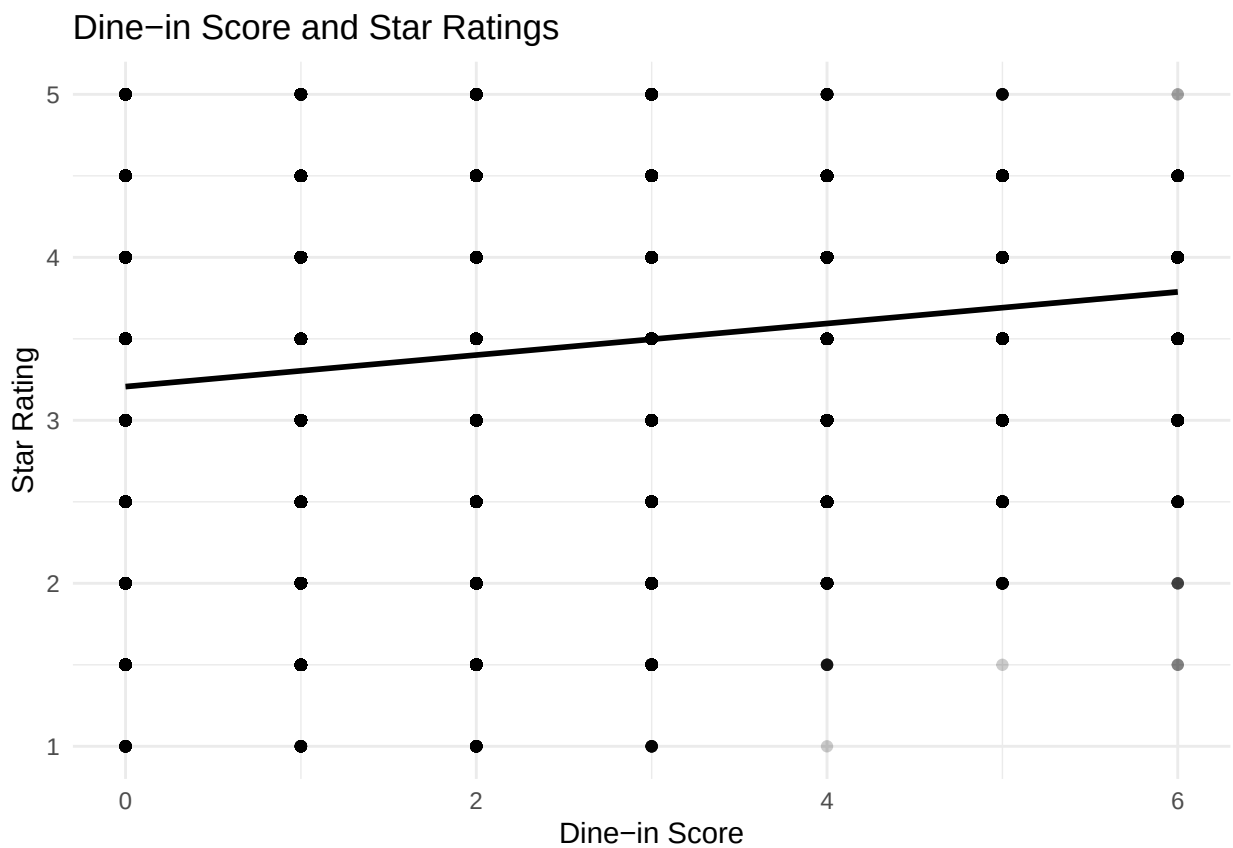
For higher-priced restaurants, delivery services are associated with higher ratings, consistent with the positive interaction effect observed in the regression analysis. Although the slope of the highest price category appears relatively flat visually, the regression results confirm a statistically significant positive interaction effect, indicating that the association between delivery services and ratings becomes more favorable at higher price levels.

Dine_in_score

```
#summary statistics
summary(dataset_restaurants$dinein_score)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   0.000   2.000   3.000   2.666   4.000   6.000
```

```
#Dataplot for dine_in_score
plot_dinein_stars <- ggplot(dataset_restaurants, aes(x = dinein_score, y = stars)) +
  geom_point(alpha = 0.2) +
  geom_smooth(method = "lm", color = "black") +
  scale_y_continuous(limits = c(1, 5)) +
  labs(
    title = "Dine-in Score and Star Ratings",
    x = "Dine-in Score",
    y = "Star Rating"
  ) +
  theme_minimal()

plot_dinein_stars
```




```r
ggsave(
  filename = here("gen", "output", "plot_dinein_stars.png"),
  plot = plot_dinein_stars,
  width = 7,
  height = 5
)
```

This figure shows the relationship between dine-in score and star ratings. The fitted regression line indicates a positive association, suggesting that restaurants offering more dine-in related services tend to receive higher customer ratings. The slope appears steeper than that observed for delivery score, indicating that dine-in experience may have a stronger relationship with customer evaluations in this dataset.

```r
#Dine_in_score summary
model_dinein <- lm(stars ~ dinein_score,
                   data = dataset_restaurants)
summary(model_dinein)
```

```
##
## Call:
## lm(formula = stars ~ dinein_score, data = dataset_restaurants)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.59414 -0.49730  0.09955  0.59955  1.79323
##
```

```
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.206765   0.008014  400.14   <2e-16 ***
## dinein_score 0.096844   0.002651   36.53   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.797 on 44482 degrees of freedom
## Multiple R-squared:  0.02912,    Adjusted R-squared:  0.0291
## F-statistic:  1334 on 1 and 44482 DF,  p-value: < 2.2e-16
```

A simple linear regression shows a statistically significant positive relationship between dine-in score and star rating (P <.001). This indicates that restaurants offering more dine-in related services tend to receive higher customer ratings.

The effect size is meaningful: across the observed range of the dine-in-score, the predicted difference in ratings is approximately 0.58 stars.

The model explains about 2.9% of the variance in star ratings, indicating that dine-in experience is a stronger predictor of ratings than delivery options in this dataset.

Moderation for dine-in score

```r
#Summary for dine_in_moderation
model_dinein_mod <- lm(stars ~ dinein_score * price_range, data = dataset_restaurants)
summary(model_dinein_mod)
```

```
##
## Call:
## lm(formula = stars ~ dinein_score * price_range, data = dataset_restaurants)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.05072 -0.54840  0.06739  0.53211  1.95160
##
## Coefficients:
##                           Estimate Std. Error t value Pr(>|t|)
## (Intercept)               2.679054   0.023404  114.47   <2e-16 ***
## dinein_score              0.206000   0.009239   22.30   <2e-16 ***
## price_range               0.369349   0.014318   25.80   <2e-16 ***
## dinein_score:price_range -0.077933   0.005085  -15.33   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7891 on 44480 degrees of freedom
## Multiple R-squared:  0.04831,    Adjusted R-squared:  0.04825
## F-statistic: 752.7 on 3 and 44480 DF,  p-value: < 2.2e-16
```

A regression model including an interaction between dine-in score and price range reveals a statistically significant moderation effect. The interaction indicates that the positive association between dine-in experience and star ratings becomes weaker as price range increases.

Simple slope calculations suggest that dine-in score is positively related to ratings for lower-priced restaurants. However, the strength of this relationship diminishes for mid-priced establishments and may become slightly negative for the higher price categories.

This pattern suggests that dine-in features are particularly valued in lower-priced restaurants, whereas for higher-priced establishments, additional dine-in services do not necessarily translate into higher customer ratings.

This finding contrasts with the moderation observed for delivery services, suggesting that the strategic importance of dine-in features differs across price segments.

```
#Data plot for dine_in_score and star ratings by price range

plot_dinein_mod <- ggplot(
  dataset_restaurants,
  aes(x = dinein_score, y = stars, color = factor(price_range))
) +
  geom_smooth(method = "lm", se = FALSE) +
  scale_y_continuous(limits = c(1, 5)) +
  labs(
    title = "Dine-in Score and Star Ratings by Price Range",
    x = "Dine-in Score",
    y = "Star Rating",
    color = "Price Range"
  ) +
  theme_minimal()

plot_dinein_mod
```

```r
ggsave(
  filename = here("gen", "output", "plot_dinein_mod.png"),
  plot = plot_dinein_mod,
  width = 7,
  height = 5
)
```

This figure illustrates the moderating role of price range on the relationship between dine-in score and star ratings. Across most price levels, dine-in score is positively associated with customer ratings. However, the strength of this relationship diminishes as price range increases.

For lower-priced restaurants, improvements in dine-in features are associated with relatively larger increases in star ratings. In contrast, for higher-priced establishments, the relationship is weaker and may approach zero.

This pattern is consistent with the negative interaction term observed in the regression analysis. Indicating that the effect of dine_in experience on ratings diminishes at higher price levels

Regression table

```r
model_delivery <- lm(
  stars ~ delivery_score,
  data = dataset_restaurants
)

model_moderation <- lm(
  stars ~ delivery_score * price_range,
  data = dataset_restaurants
)

model_dinein <- lm(
  stars ~ dinein_score,
  data = dataset_restaurants
)

model_dinein_mod <- lm(
  stars ~ dinein_score * price_range,
  data = dataset_restaurants
)
stargazer(
  model_delivery,
  model_moderation,
  model_dinein,
  model_dinein_mod,
  type = "text",
  title = "Regression Results",
  dep.var.labels = "Star Rating",
  column.labels = c(
    "Delivery",
    "Delivery x Price",
    "Dine-in",
    "Dine-in x Price"
  ),
  digits = 3
)
```

```
##
## Regression Results
## ==========================================================================================
##                                                                  Dependent variable:
##                             ---------------------------------------------------------------------------
##                                                                       Star Rating
##                                   Delivery               Delivery x Price                Dine-in
##                                     (1)                        (2)                          (3)
## ------------------------------------------------------------------------------------------
## delivery_score                    -0.063***                  -0.316***
##                                    (0.004)                    (0.011)
##
## price_range                                                  -0.066***
##                                                              (0.013)
##
## delivery_score:price_range                                    0.168***
##                                                              (0.006)
##
## dinein_score:price_range
##
##
## dinein_score                                                                          0.097***
##                                                                                       (0.003)
##
## Constant                           3.587***                   3.674***                   3.207***
##                                    (0.009)                    (0.024)                    (0.008)
##
## ------------------------------------------------------------------------------------------
## Observations                        44,484                     44,484                     44,484
## R2                                  0.005                      0.047                      0.029
## Adjusted R2                         0.005                      0.047                      0.029
## Residual Std. Error            0.807 (df = 44482)         0.790 (df = 44480)          0.797 (df = 44
## F Statistic                240.985*** (df = 1; 44482) 727.898*** (df = 3; 44480) 1,334.399*** (df =
## ==========================================================================================
## Note:
```